

MONTICELLO, IA, USA

WMO: 725475

Lat: 42.224N

Lon: 91.166W

Elev: 259

StdP: 98.25

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04953

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.5	(c) -19.5	(d) -27.0	(e) 0.3	(f) -22.4	(g) -23.6	(h) 0.5	(i) -19.1	(j) 12.6	(k) -7.7	(l) 11.4	(m) -7.2	(n) 2.0	(o) 290	(p) 0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	32.5	24.8	31.2	24.1	29.8	23.3	26.4	30.5	25.4	29.6	24.3	28.4	4.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.9	29.2	23.9	19.4	28.2	22.8	18.0	27.0	84.1	30.4	78.6	29.7	74.7	28.4	33.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.0	(b) 9.3	(c) 8.3	(e) -26.9	(f) 34.4	(g) 3.8	(h) 1.9	(i) -29.6	(j) 35.8	(k) -31.8	(l) 37.0	(m) -34.0	(n) 38.1	(o) -36.7	(p) 39.5		
(a) 11.0	(b) 9.3	(c) 8.3	(e) -27.7	(f) 28.0	(g) 4.1	(h) 1.1	(i) -30.7	(j) 28.8	(k) -33.1	(l) 29.4	(m) -35.4	(n) 30.0	(o) -38.4	(p) 30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-6.3	-4.5	2.9	9.1	15.9	21.1	22.7	21.6	17.5	10.5	3.2	-3.5		
	DBStd	11.56	6.93	7.16	6.47	5.37	4.73	3.45	3.09	2.97	4.49	5.20	6.09	6.54		
	HDD10.0	1928	504	406	235	78	6	0	0	0	3	58	218	421		
	HDD18.3	3737	763	638	479	278	107	13	3	6	67	249	454	678		
	CDD10.0	1659	0	1	14	52	188	334	393	359	228	73	14	1		
	CDD18.3	425	0	0	1	2	31	97	138	107	43	5	2	0		
	CDH23.3	4075	0	0	7	54	315	911	1333	939	426	56	34	0		
	CDH26.7	1300	0	0	0	9	85	302	464	284	127	10	19	0		
(14) Wind	WSAvg	3.4	3.9	3.9	4.1	4.4	3.9	3.1	2.2	2.1	2.5	3.4	3.7	3.6		
(15) Precipitation	PrecAvg	966	32	36	56	95	119	143	124	111	84	70	57	41		
	PrecMax	1323	62	97	141	196	222	236	312	222	160	188	164	92		
	PrecMin	688	8	2	11	30	17	30	15	10	18	12	4	6		
	PrecStd	159	14	24	37	41	50	57	72	56	36	50	44	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	15.9	24.0	27.8	32.0	33.5	36.1	33.1	32.4	27.9	28.7	15.2		
		MCWB	6.9	10.5	16.6	17.7	22.4	24.3	25.2	24.8	24.1	19.0	25.4	12.4		
	2%	DB	7.2	10.3	19.8	25.0	28.9	32.1	32.5	31.3	30.1	24.9	18.8	10.1		
		MCWB	4.7	7.4	14.0	16.2	20.1	23.8	25.9	24.7	22.8	16.8	13.7	7.5		
	5%	DB	4.0	7.2	16.1	22.1	27.1	30.1	31.1	29.8	27.7	22.3	15.1	7.4		
		MCWB	2.3	4.5	12.2	14.2	18.9	22.6	24.9	24.2	20.9	16.3	10.9	5.4		
	10%	DB	2.4	4.0	12.5	18.8	24.8	28.0	29.0	27.8	26.0	19.8	12.4	4.7		
		MCWB	1.1	2.1	8.8	12.7	17.8	21.4	23.8	22.6	19.4	15.0	9.1	2.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	11.2	18.0	19.7	23.8	26.8	28.4	27.2	25.5	22.1	23.7	13.4		
		MCDB	9.1	15.0	22.6	26.3	29.4	31.2	32.5	31.5	30.4	26.6	28.6	15.8		
	2%	WB	4.6	7.6	15.3	17.4	21.9	25.2	26.7	25.9	24.2	19.2	14.4	8.0		
		MCDB	7.0	10.4	17.9	22.7	27.1	30.0	30.5	29.9	28.5	22.1	17.8	9.1		
	5%	WB	2.6	4.6	12.2	15.4	20.3	23.9	25.6	24.6	22.6	17.2	11.8	5.2		
		MCDB	4.3	6.8	16.0	20.8	24.8	28.3	29.7	28.4	26.0	20.5	14.7	7.1		
	10%	WB	1.1	2.5	9.1	13.3	19.0	22.7	24.5	23.6	20.9	15.2	9.0	2.9		
		MCDB	2.2	4.2	12.3	18.4	23.3	27.1	28.6	27.0	23.8	19.0	12.3	4.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.3	10.2	12.0	11.8	11.1	11.2	11.4	13.0	12.0	10.1	8.3		
		MCDBR	10.6	11.5	14.8	17.0	13.9	13.0	12.2	12.6	14.4	15.3	14.4	10.5		
	5% WB	MCWBR	8.6	8.8	9.7	9.7	7.3	6.8	7.1	7.8	7.9	8.8	10.2	8.6		
		MCWBR	9.9	11.2	13.7	14.5	12.0	11.2	11.2	10.8	11.9	12.3	13.5	9.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.284	0.299	0.338	0.377	0.409	0.427	0.438	0.415	0.386	0.339	0.297	0.281		
		taud	2.391	2.357	2.365	2.279	2.254	2.245	2.206	2.274	2.323	2.456	2.497	2.430		
	Ebn at Noon	Ebn at Noon	865	906	907	891	867	849	835	844	843	850	846	837		
		Edh at Noon	79	96	109	128	135	136	140	127	112	87	71	70		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.79	2.79	3.77	4.66	5.51	6.11	6.13	5.38	4.49	2.93	1.95	1.43		
	RadStd	RadStd	0.20	0.28	0.37	0.38	0.39	0.53	0.40	0.35	0.36	0.36	0.15	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page